Mock Midterm Worked Solutions

Machine Learning in Engineering Science

Fall 2026 | Revised 9 November 2026 | NCKU + LAiMM

Worked solutions and scoring guidance | Public Mock practice

Attempt the 75-minute paper before reading. All numerical cases are synthetic teaching examples; this Mock has no course-grade weight.

Award points for the requested decision, method and numerical result. Accept equivalent correct reasoning and exact/reduced fractions. A correct fraction earns full numeric credit without an unnecessary decimal or duplicated calculation. When arithmetic is wrong, retain the indicated method credit. Carry a single isolated arithmetic error forward without repeated penalties; cap each subpart at its stated maximum. Do not award an unasked extra claim or require an unrelated limitation. Answer-length guidance is not a separate grading criterion.

Activity	Points	Minutes
Read and plan	0	3
1 Frame a vibration prediction experiment	20	12
2 Evaluate alarms under asymmetric costs	15	10
3 Relate variance retention to prediction	10	7
4 Reason about a causal sequence model	25	19
5 Diagnose variation and a changed deployment	20	13
6 Check predictions before claiming reproduction	10	7
Final check	0	4
Total	100	75

Timing is a workload design estimate, not a measured guarantee for every student. The paper retains six questions and 100 points, uses compact answer formats and includes four final-check minutes. The oral uses one short case plus one fixed follow-up in 120 seconds.

Question 1 Frame a vibration prediction experiment

20 points | Week 1 to 3 | Judgments: target, split, baseline, evaluation, limitations

Part a 4 points

1 point One pump run is a row; runs within a pump can be dependent.

1 point Target is vibration RMS ten seconds later in mm/s.

1 point The prediction is issued at run start with then-available inputs.

1 point This is supervised regression.

Part b 6 points

1 point Identify shared pump identities under a random row split.

1 point Keep whole pump IDs disjoint for unseen-pump deployment.

1 point Identify fitting the scaler before splitting as held-out information use.

1 point Fit the scaler on training rows only; apply that fitted transformation elsewhere.

1 point Use validation to choose settings.

1 point Freeze before testing on the untouched pump groups.

Part c 4 points

1 point Prediction = 0.5 + 0.05(30) + 0.2(2) = 2.4 mm/s.

1 point Signed residual = 2.4 - 3 = -0.6 mm/s.

1 point Absolute error = 0.6 mm/s.

1 point Squared error = 0.36 (mm/s) squared.

Part d 6 points

2 points Use the training-target mean as the squared-loss-optimal constant. Accept another explicitly training-only constant as a justified comparator; do not claim that a median must minimize RMSE.

2 points Compare on the same held-out pump groups, metric and permitted information.

1 point Ridge penalizes coefficient size and trades fit against regularization; a larger penalty does not guarantee a smaller absolute value for every individual coefficient.

1 point Regularization does not repair group overlap or globally fitted preprocessing.

Marking cautions

A different random seed does not separate pump identities.

Question 2 Evaluate alarms under asymmetric costs

15 points | Week 5 | Judgments: evaluation, limitations

Part a 4 points

1 point TP = 12.

1 point FN = 8.

1 point FP = 6.

1 point TN = 54.

Part b 6 points

1.5 points Precision =12/(12+6) =2/3 ≈0.667. Award 0.5 for the correct conditional denominator and 1 for the resulting fraction/value.

1.5 points Recall = sensitivity =12/(12+8) =0.600. Award 0.5 for the correct conditional denominator and 1 for the fraction/value.

1.5 points Specificity =54/(54+6) =0.900. Award 0.5 for the correct conditional denominator and 1 for the fraction/value.

1.5 points F1 =24/(24+6+8) =12/19 ≈0.632. Award 0.5 for the correct substitution and 1 for the fraction/value. A correct fraction earns full numeric credit; no decimal is required.

Part c 5 points

1 point Threshold 0.4 cost =8×2+20 =36.

1 point Threshold 0.7 cost =8×8+6 =70.

1 point Choose 0.4 using validation only.

1 point Freeze the setting before final test evaluation.

1 point No: test predictions at 0.7 do not determine the score at 0.4. New predictions at the frozen setting are needed. Do not use test scores to choose it.

Marking cautions

Keep validation and test evidence separate even when class counts coincide.

Question 3 Relate variance retention to prediction

10 points | Week 6 | Judgments: evaluation, limitations

Part a 2 points

1 point Total input variance =1.6+0.4 =2.

1 point Retained fraction =1.6/2 =0.80 or 80%.

Part b 2 points

1 point PC1 score = 2/sqrt(2) = sqrt(2).

1 point PC1-only reconstruction = (1, 1).

Part c 2 points

1 point The discarded residual is z−reconstruction =(1,−1), in the feature-difference direction.

1 point That direction can carry label information;80% retained input variance does not guarantee retention of the fault signal.

Part d 4 points

1 point The coordinate differences become 1000 times larger.

1 point Their squared-distance contribution becomes 10^6 times larger, changing the clustering objective. Actual cluster assignments may or may not change.

1 point Cluster IDs are arbitrary labels for geometric groups.

1 point Physical fault interpretation requires independent evidence or validation.

Marking cautions

The reconstruction is not (2, 0); the PC2 component has been dropped.

Question 4 Reason about a causal sequence model

25 points | Week 7 | Judgments: target, split, evaluation, limitations

Part a 4 points

1 point Input [1, 3, 2, 5].

1 point Target x4 = 4.

1 point x4 is future and unavailable at time 3.

1 point Assign targets chronologically and construct all features from past/available information only; historical context can be used when legitimately known.

Part b 6 points

2 points Cell = 0.6(1)+0.5(0.8) = 1.0.

2 points Hidden = 0.25(0.7616) = 0.1904.

2 points Closing only the output gate gives hidden = 0, cell = 1 at this step. It does not erase memory, and later dynamics need not be unchanged.

Part c 8 points

1 point Exclude key 2: valid exponentials [2, 0, 3]; the masked logit is−infinity.

2 points Valid denominator 5 gives weights [2/5, 0, 3/5].

1 point Use the weighted sum of valid values.

2 points Context =(2/5)[5, 0]+(3/5)[0, 5] =[2, 3].

1 point Zeroing after ordinary softmax gives [2/6, 0, 3/6], sum 5/6.

1 point Renormalize over valid keys. Zeroing AND renormalizing is also correct; zeroing alone is not.

Part d 7 points

3 points Positions 0–3 agree (1). They cannot access the changed suffix (1), and their available prefix/other inputs are identical (1). Later logits need not differ.

2 points Teacher forcing supplies all shifted input tokens, permitting masked parallel training (1). Ordinary generation needs previous generated tokens for the next step (1).

2 points Explicit positions give location/order/distance cues (1); causal masking restricts future access (1). Some causal models infer implicit positions; do not require an order-blind-without-explicit-positions claim.

Marking cautions

The middle key, not the last key, is padding here.

Unchanged prefixes must agree; changed suffix logits need not differ. Masking and explicit positions have distinct roles.

Question 5 Diagnose variation and a changed deployment

20 points | Week 9 | Judgments: evaluation, limitations

Part a 6 points

2 points Bias = 1.6 - 2.0 = -0.4; squared bias = 0.16.

2 points Affine MSE = 0.16 + 0.01 + 0.25 = 0.42.

2 points Nonlinear MSE = 0 + 0.08 + 0.25 = 0.33.

Part b 4 points

1 point Affine refit variance falls from 0.25 to 0.01.

1 point Mean prediction remains 1.6, so the estimated bias at this input remains−0.4.

1 point A persistent curved mean residual can suggest model-form misspecification; more rows do not enlarge the affine hypothesis class. The curve alone is not a unique proof of the mechanism.

1 point The stated independent noise floor remains 0.25 under the same available information. More training rows do not remove it.

Part c 4 points

2 points A is input/covariate shift with fixed P(Y given X).

2 points B changes P(Y given X); unchanged marginal inputs would not prevent this relationship shift.

Part d 6 points

2 points Test MAE = 0.2(3)+0.8(0.5) = 1.0.

2 points Reweighted MAE = 0.4(3)+0.6(0.5) = 1.5.

1 point The group-conditional mean absolute errors must remain stable.

1 point Changes within a group, target-response/noise changes or subgroup sampling uncertainty can invalidate the estimate. It is not direct deployment validation.

Marking cautions

MSE and RMSE are not interchangeable in the decomposition.

Question 6 Check predictions before claiming reproduction

10 points | Week 10 | Judgments: split, evaluation, limitations

Part a 3 points

1 point Check unique IDs and equal ID sets; reject duplicates/missing/extra IDs.

1 point Align by ID rather than row index.

1 point Aligned maximum absolute difference = 0 for each model.

Part b 2 points

1 point An equal rounded aggregate loss can conceal different predictions on individual rows.

1 point Compare every ID-aligned prediction for both frozen models under the predeclared tolerance.

Part c 2 points

1 point Reject the changed unit contract before fitting.

1 point Unchanged numbers with different units have a different interpretation. Verify source units; a justified rpm↔revolutions/second conversion needs a factor 60 and explicit reviewed implementation.

Part d 3 points

1 point Hashes support byte consistency relative to the reference; aligned numerical agreement supports the stated computational replay. Either correctly bounded claim earns this point.

1 point Give one unsupported stronger claim: authenticated authorship, independent human/environment verification, physical validity, or generalization. One explicit valid example is enough.

1 point Record a relevant Python,NumPy,scikit-learn or OS/runtime version. A version name is sufficient; no actual version number must be memorized.

Marking cautions

A data checksum is not a physical validation certificate.

Two minute oral administration and rubric

Within the slot	Action
0–15 seconds	Read context and main question only. Keep the follow-up covered.
15–75 seconds	Answer the main question: decision, supplied clue, next step and a short limit.
75–105 seconds	Examiner reads the fixed follow-up; student answers within the same 30 seconds.
105–120 seconds	Close and record the score.

Criterion	Points	Evidence
Decision and mechanism	12	6 central decision +6 correct mechanism.
Evidence and next step or limit	10	4 supplied evidence +3 justified next step +3 concrete scope limit.
Counterfactual reasoning	8	4 updated conclusion +4 correct explanation; a yes/no alone cannot earn all 8.

Assess the response across the whole slot; a concise limitation may appear in the follow-up. Do not require a fourth topic beyond the printed prompts. Equivalent reasoning earns the same credit. Keep examiner clarification neutral and the follow-up fixed. Seat changes, if needed, require separate scheduling; they must not selectively reduce candidate answering time.

To rehearse the confirmed midterm weighting: midterm practice score =0.70×written raw score out of 100 + oral raw score out of 30. For example,80 written and 24 oral gives 80/100. This does not set the midterm share of the semester grade.

M1 Selection after test

A student tries five Ridge alphas, selects the lowest test RMSE, and reports it as an unbiased final estimate.

Main: Identify the problem. Use the evidence to propose a next step and state one limitation.

Fixed follow-up: Would a fixed random seed make that estimate valid? Why?

Criterion	Points	Case-specific anchors
Decision and mechanism	12	6: identify test-driven selection;6: explain why the selected minimum is not untouched final evidence.
Evidence and next step or limit	10	4: use the five choices/same-test evidence;3: validation,freeze and genuinely untouched evaluation;3: state a concrete scope limit.
Counterfactual reasoning	8	4: reject the claim;4: explain repeatability versus information use.

Concise full-credit example

Main response: The test scores selected among five models, so their minimum is a development result rather than untouched final evidence. Select on training/validation, freeze the comparison and use a genuinely untouched final holdout. One holdout still does not guarantee performance on every future population.

Follow-up response: No. A seed can repeat the procedure; it does not undo the test information used for selection.

This example covers 12+10+8=30 points. It is a reasoning example, not a required script. A different correct, concise explanation can earn the same marks.

Partial-credit example

Main: “They used test scores to choose alpha. Use validation next.” Follow-up: “No.”

Illustrative score:9/12 (6 decision +3 partial mechanism),5/10 (4 supplied clue +1 incomplete next step +0 explicit limit),4/8 (correct conclusion, no explanation),total 18/30. The next step omits freezing and genuinely untouched final evidence. Award from actual reasoning rather than the sample label.

M2 A unit label in a replay

An airflow package requires m/s. Its copied contract says km/h with unchanged numbers. The loader rejects it before fitting.

Main: Is this useful rejection or model failure? Give an evidence-based next step and one limitation.

Fixed follow-up: Does relabeling it to m/s prove the original units were correct? Why?

Criterion	Points	Case-specific anchors
Decision and mechanism	12	6: identify useful pre-fit contract enforcement;6: explain the numerical-unit interpretation mismatch.
Evidence and next step or limit	10	4: cite unchanged values/different unit labels;3: verify source units,review correction and retest;3: distinguish computation from physical meaning.
Counterfactual reasoning	8	4: answer no;4: explain that a shared unit error can reproduce and provenance is still required.

Concise full-credit example

Main response: This is useful input-contract enforcement, not a fitting failure. The changed unit label changes what the same numbers mean. Preserve the rejected package, verify source units and make a justified correction or conversion explicit before retesting. Passing replay alone does not validate the physical measurements.

Follow-up response: No. Matching predictions can reproduce the same encoding error. Verify the original measurement units and provenance.

This example covers 12+10+8=30 points. It is a reasoning example, not a required script. A different correct, concise explanation can earn the same marks.

Workload and marking notes

Written budget:3 minutes initial reading +68 minutes across six questions +4 minutes checking =75 minutes. Question times are 12, 10, 7, 19, 13, 7 minutes. Use the short answer formats; the exam does not ask for essays or executable programs.

The timing estimate accounts for question reading, handwriting, calculation and mechanism selection. No student timed pilot is claimed. If a representative rehearsal exceeds the budget, record the specific subpart rather than silently extending the formal exam or removing a learning objective.

For the two-minute oral, the context+main cards contain fewer than 40 words each. Full-credit model responses are intentionally short. The follow-up has at most 16 words and its reading is included in the 30-second follow-up period. A150-word/minute card-reading pace and 90-word/minute speaking pace are explicit planning assumptions, not a fluency scoring rule.

Technical clarification

In Question 4d, explicit positions and causal masking have different roles. Do not insist that a causal model without explicit positional encodings has no order information: some causal models learn implicit position information. The requested prefix-invariance check holds only under the stated deterministic setup and unchanged nonsequence inputs.

Primary references used for the review

Training-only preprocessing: https://scikit-learn.org/stable/common_pitfalls.html

LSTM cell and hidden output equations: https://docs.pytorch.org/docs/2.9/generated/torch.nn.LSTM.html

Masked decoding and explicit positions: https://arxiv.org/abs/1706.03762

Implicit position in causal language models: https://arxiv.org/abs/2203.16634